JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PRASANNA CORPORATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

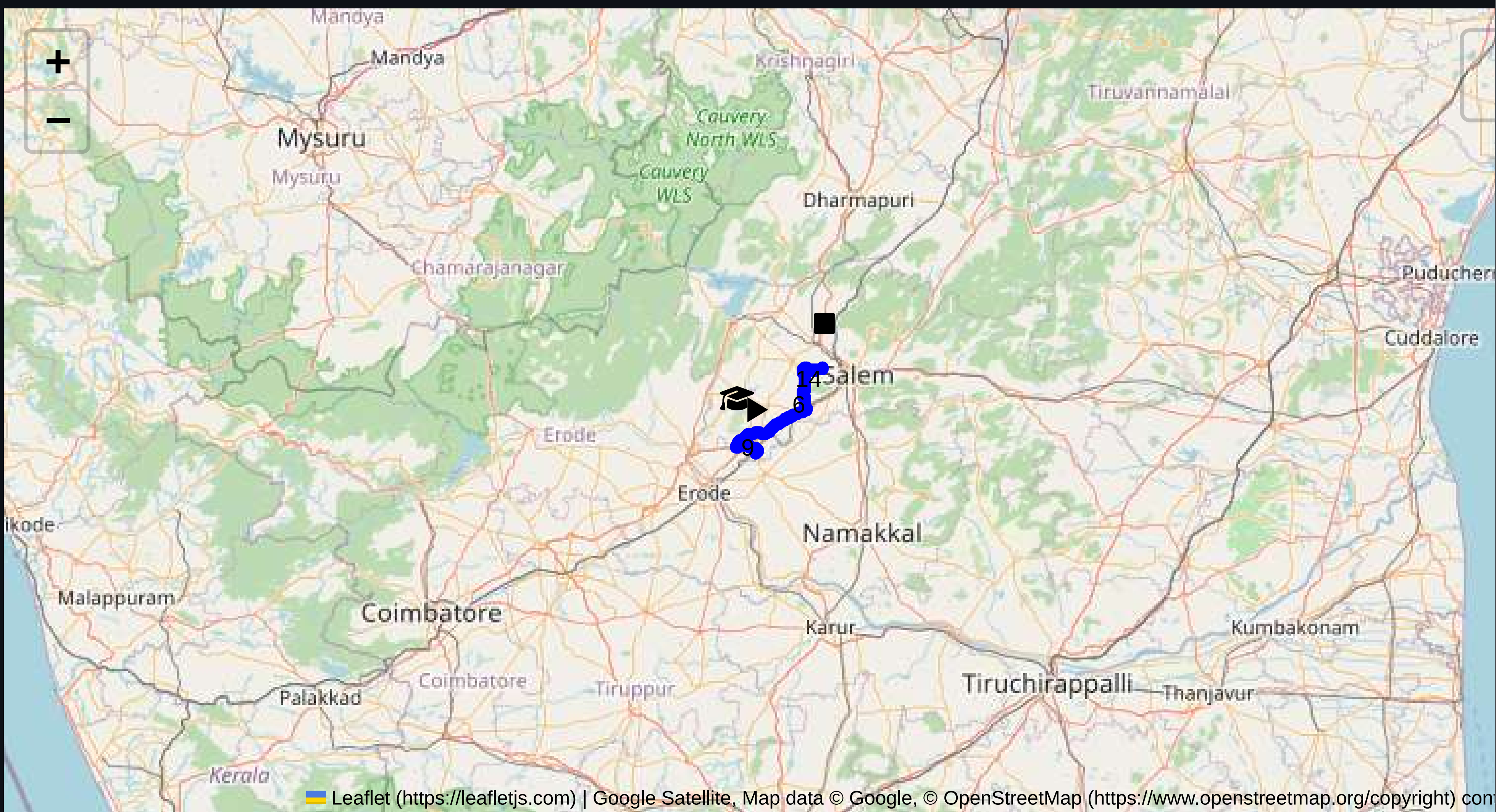
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 53.72 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.671512, 78.057213)



Welcome to the Journey Risk Management Study

To evaluate the safety of the route from Sangagiri to the Salem Steel Plant 1st Gate, taking into consideration major points as requested, I will provide a comprehensive analysis using available data and context.

1. Overview of the Route Map

- The route from Sangagiri to the Salem Steel Plant 1st Gate is about 53.72 kilometers long.
- This involves driving through diverse terrains including rural routes and more urbanized areas as you approach Salem.
- Key waypoints include passing through Padaiveedu and annapuranimass in Salem.

2. Typical Weather Conditions and Potential Weather-related Hazards

- Tamil Nadu generally experiences a tropical climate. Expect high temperatures and humidity.
- Monsoon season (October to December) can result in heavy rains, causing slippery roads and possible flooding, especially in low-lying areas.
- During summer months, high temperatures may affect vehicle performance.

3. Analysis of Traffic Patterns

- Traffic congestion is notably high during morning and evening peak hours around Salem.
- Rural segments may see less traffic but be wary of slow-moving vehicles and livestock.
- Check local traffic advisories for any roadwork or accidents.

4. Assessment of Road Quality and Infrastructure

- Roads in rural areas may have uneven surfaces and require cautious driving due to potholes and narrow lanes.
- Urban sections feature better-maintained roads, though these can be crowded.
- No significant expressways—expect standard two-lane highways.

5. Suggestions for Alternative Routes for Emergencies

- NH44 can be considered if the primary route is inaccessible, keeping in mind it runs parallel but may lengthen travel.
- Local roads bypassing Salem city center are useful to avoid congestion during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Tamil Nadu regulations enforce strict timing windows for heavy vehicles in urban areas to minimize traffic disruptions.
- Ensure proper documentation of hazardous materials, including placarding and emergency contact details, as per legal requirements.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Historical data indicates sporadic incidents involving overturns on sharp turns and mishaps due to hydroplaning during rains.
- Incidents in Salem often relate to vehicular congestion rather than specific hazardous material mishaps.

8. Environmental Considerations and Sensitive Areas

- The route may pass near agricultural lands where contamination could have severe impacts.
- Noise and emissions from heavy vehicles in residential areas should be minimized.

9. Analysis of Communication Coverage

- Rural segments might suffer from occasional coverage dropouts, particularly in less populated areas.
- The closer to Salem, the more reliable the cellular network coverage.

10. Estimated Emergency Response Times

- Urban locations near Salem have quicker response times due to proximity to facilities.
- Rural areas might see delayed response times, ranging from 20 to 40 minutes depending on location and traffic condition.

12. Overall Summary of Risk Assessment

- **Weather**: Monitor conditions during the monsoon season and plan accordingly.
- **Traffic**: Be aware of congestion in Salem, favoring non-peak times for travel.
- **Road Quality**: Exercise caution on rural roads with potential hazards like potholes.
- **Regulations & Emergency Response**: Ensure all regulatory compliance and be prepared for variation in emergency response times.
- **Safety Protocols**: Regularly review safety protocols related to hazardous materials. Ensure contingency plans for communication failures and identify suitable detours.

This analysis provides a broad overview to facilitate safer transport on the specified route, aiming to prepare drivers for possible risks and ensure adherence to safety regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	Medium	11.45353, 77.85697	30 KM/Hr
4	Turn	Medium	11.45839, 77.85143	30 KM/Hr
5	Turn	Medium	11.46307, 77.84941	30 KM/Hr
6	Turn	Medium	11.46316, 77.84921	30 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
9	Turn	High	11.55839, 78.01288	15 KM/Hr
10	Turn	Medium	11.60526, 78.00451	30 KM/Hr
11	Turn	Medium	11.61685, 78.00830	30 KM/Hr
12	Turn	High	11.62660, 78.00986	15 KM/Hr
13	Turn	High	11.62692, 78.00883	15 KM/Hr
14	Turn	Medium	11.64031, 78.00923	30 KM/Hr
15	Turn	Medium	11.64874, 78.00717	30 KM/Hr
16	Turn	Medium	11.66065, 78.00106	30 KM/Hr
17	Blind Spot	Blind Spot	11.66821, 78.00077	10 KM/Hr
18	Turn	High	11.66762, 78.00863	15 KM/Hr
19	Blind Spot	Blind Spot	11.67168, 78.00922	10 KM/Hr
20	Turn	High	11.67094, 78.05691	15 KM/Hr
21	Turn	Medium	11.67120, 78.05691	30 KM/Hr
22	Turn	Medium	11.67126, 78.05696	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium
4	hospital	Subam Hospital	11.6069532, 78.0042593	30 km/h	Medium
5	hospital	SR Hospital	11.6075985, 78.0070051	30 km/h	Medium
6	hospital	Nesam Medical Centre and Hospital	11.6078647, 78.0073456	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45353, 77.85697

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45839, 77.85143

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46316, 77.84921

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60526, 78.00451

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62660, 78.00986

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62692, 78.00883

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64031, 78.00923

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64874, 78.00717

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.66821, 78.00077

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66762, 78.00863

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67168, 78.00922

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67094, 78.05691

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67120, 78.05691

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67126, 78.05696